

Adam “Jack” Fraser

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Work eligibility: Valid New Zealand student visa with work rights

PROFILE

Postgraduate Computer Science student at the University of Auckland with research and industry experience in computer vision and deep learning. Currently fine-tuning and evaluating a locally-run vision-language model for a University research project. Previously designed a TensorFlow/Keras image classifier for a South Carolina state government project and benchmarked it against AWS Rekognition and SageMaker. Several years of professional software and systems engineering before returning to study. Comfortable working independently in Python-based research environments, from model design and training through to evaluation and technical write-up.

Leads from the front while maintaining that a rising tide lifts all boats: the goal is an environment where talent is directed at interesting, challenging and valuable problems, hard work is rewarded, and progress is steady. Attention to detail and an eye for audit points and vulnerabilities raise the quality and security of whatever the team ships.

EDUCATION

Postgraduate Diploma in Science, Computer Science — University of Auckland, Auckland, NZ 2026 – Present

First semester of study in New Zealand; no University of Auckland transcript is available yet.

Bachelor of Science, Computer Science — College of Charleston, Charleston, SC, USA 2015 – Dec 2019

Relevant coursework: Advanced Algorithms (A), Software Engineering & Practicum (A), Programming Language Concepts (A), Database Concepts (A), Linear Algebra (A), Statistical Methods, Computer & Network Security. Highly Distinguished standing in final two semesters.

RELEVANT RESEARCH & TECHNICAL PROJECTS

Local Vision-Language Model Evaluation for Anomaly Detection — University of Auckland, COMPSCI 742 2026

Group research project on how input representation (structured data, text, images) affects a transformer vision-language model fine-tuned on top of open base weights.

- Evaluated candidate open-weight models and selected a vision-language model that could be trained and run entirely on local hardware.
- Designed and built a custom Python testing framework to benchmark base and fine-tuned variants across structured, text, image and multimodal inputs.
- Ran logistic regression and random forest baselines on the same held-out data for direct comparison.

Computer Vision for Fish-Egg Classification — SC Department of Administration, Columbia, SC Sep – Dec 2025

Internal technical evaluation supporting the Dept of Natural Resources' fish-restocking program, part of South Carolina's statewide AI initiative. Conventional (non-LLM) deep learning.

- Designed, trained, and tested a customised ReLU neural network in TensorFlow/Keras, with a final translation layer mapping learned features to classifications, to classify and predict striped bass egg hatching readiness from image data.
- Compared training with and without image preprocessing; modified the architecture to give scalar rather than categorical outputs.
- Ran training and inference on State of South Carolina datacentre compute; peak training load shifted datacentre CPU utilisation by around 25%.
- Authored the feasibility, cost and effectiveness report benchmarking the custom model against AWS Rekognition and AWS SageMaker, co-written with a mentored intern.
- Backup administrator on the state's OpenAI Enterprise pilot, working with the AI Center of Excellence on capability and requirements assessment.

TECHNICAL SKILLS

Languages & ML frameworks: Python, PyTorch, TensorFlow/Keras, Hugging Face Transformers, PEFT/QLoRA, TypeScript, SQL

Computer vision & AI: custom neural-network design and training (non-LLM and transformer/VL), vision-language models, image classification, model fine-tuning and evaluation, AWS Rekognition, AWS SageMaker

Tooling: Node.js, Git, Azure DevOps (CI/CD), Linux

OTHER PROFESSIONAL EXPERIENCE

Systems Administrator, promoted to Systems Engineer I — SC Dept of Administration, Columbia, SC 2023 – 2026

Server Infrastructure / Data Center Infrastructure team. Rated "Exceptional" in the 2026 annual performance review.

Development

- Designed and deployed "Omnitool", an internal Node.js/TypeScript web app giving helpdesk staff a unified interface for user verification, password management, two-factor settings, and account linking.
- Architected the Azure DevOps CI/CD pipeline (regression tests, build, artifacts, canary deployment) and hardened the codebase with strict TypeScript typing, input sanitisation, API rate limiting and unit/integration tests.
- Requested and mentored multiple interns through the OTIS internship programme, tailoring projects to each intern's strengths and teaching agile practice, backlog planning, branch management and testing.
- Patched bugs and severe security vulnerabilities in scripts and web apps written by other authors.

Systems & security

- Architected low-downtime upgrades of sensitive user-monitoring systems via a hot-swap IP/DNS approach.
- Migrated per-server certificates and firewall exceptions to a NetScaler reverse proxy, simplifying network configuration and tightening subnet security.
- Built an internal CA certificate-expiry alerting system that raises ServiceNow tickets automatically.
- Administered vCenter VMs, server provisioning and upgrades, and Linux automated patching.

Identity & access management

- Extended Quest Password Manager to deliver multi-tenant DUO two-factor authentication on a product without native 2FA.
- Designed audit and enforcement scripts maintaining state policy across 100,000+ accounts, 10+ customer organisations, and 3 AD domains.
- Built a customer-facing ServiceNow-to-AD integration automating secure account provisioning, replacing 180+ manual account tasks within months of launch, and implemented the SAP OpenHub HR-data sync into AD to replace the legacy process.

Independent Research & Development — Self-directed 2020 – 2022

- Built neural networks from scratch and in TensorFlow, including a genetic-algorithm architecture search, to predict market movements; studied quantitative and options-trading strategies. Research ended when the GPU caught fire.

Software Developer — Charleston County Public Works, Finance Office, Charleston, SC Aug 2019 – Mar 2020

- Built a Node.js/Electron application suite (PCARD, travel requests) with installer and auto-update handling, part-time during final undergraduate year.

Summer Internships — Charleston County Public Works, Charleston, SC 2018 – 2019

- Rebuilt the county stormwater-fee calculation as a standalone application; built and distributed a Python/Tkinter/SQL PCARD reconciliation tool; prototyped VBA-based travel-request routing.